جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
نزوى
Nizwa

**INFORMATION TECHNOLOGY DEPARTMENT**

# Final Examination
# Spring, AY 2022/2023
# CSPG1205 – Introduction to Programming
# Set A

**Date:** Sun, June 4, 2023 **Time:** 9:00 AM – 11:00 AM

| Max Marks | 40 | Time Allocation | 2 Hours |
|---|---|---|---|

| Student ID | | Student Name | |
|---|---|---|---|
| Section | | Level | Diploma First Year |

**Instructions to the Students**

1. Answer the questions in the space provided by using **blue or black pen** only. You will obtain a zero (0) if you use a pencil for all question.
2. Use of mobile phones and any electronic gadgets inside the examination hall is not allowed.
3. ***Copying, Cheating, and any kind of malpractices on the examination are strictly prohibited.*** Action will be taken as per the "Article 68 of UTAS Academic Regulations".
4. Students should bring their own pencils, pens, erasers, rulers, and any other tools required for the exam.
5. The exam has a total of **15 non optional questions**.
6. **Use of standard calculators is not allowed for this exam.**

| Section | Allotted Marks | Obtained Marks |
|---|---|---|
| Section A | **10** | |
| Section B | **6** | |
| Section C | **6** | |
| Section D | **6** | |
| Section E | **12** | |
| Total Marks | **40** | |

| Marked By | Checked/Re-marked by |
|---|---|
| | |

**Section A - Multiple Choice Questions**
**(10 x 1 = 10 Marks)**

| Section A: | / 10 |
|---|---|

1) What is the output produced by the following code?

```
def myFun( x , y = 5 , z = 10):
    print( x , end = " " )
    print( y , end = " " )
    print( z , end = " " )

#call myFun()
myFun(10,10)
```

a) 10 5 10
b) 10 10 10
c) 10 15 10
d) 10 15 20

2) What is the output produced by the following code?

```
def fun(n):
    if n <= 10:
        print(n, end=" ")
        return fun(n * 2)
    else:
        return 1
#call fun()
fun(1)
```

a) 2 4 8
b) 1 2 3 4 5
c) 1 2 4 8
d) 1 2 4 8 10

3) Select the statement to insert an element with value 30 before 43 in the following list.

```
marks = [20,36,43,52,22,35]
```

a) marks.insert(2, 30)
b) marks.insert(1, 30)
c) marks.append(1, 30)
d) marks.append(2, 30)

4) Which of the following is **incorrect** according to the following tuple declaration?

```
age = ( 22 , 31 , 52 , 49 , 34 )
```

a) age = age + (33, 35)
b) age[0] = 25
c) print(age[0])
d) Maximum = max(age)

5) What is the output produced by the following code?

```
s1 = {6, 4, 5, 8, 7, 11}
s2 = {5, 2, 3, 6, 9, 4}
print(s1.intersection(s2))
```

a) { 7, 8, 11 }
b) { 4, 5, 6 }
c) { 2, 3, 9 }
d) { 6, 4, 5, 8, 7, 11, 2, 3, 9 }

6) What is the output produced by the following code?

```
d = {3: 'd', 2: 'c', 1: 'b', 0: 'a'}
print(d.popitem())
```

a) (3, 'd')
b) [3, 'd']
c) 0, 'a'
d) (0, 'a')

7) Which of the following slices will extract the following substring “PYTHON”?

```
_str = "INTRODUCTIONtoPYTHON"
```

a. print(_str[15:20])
b. print(_str[15:])
c. print(_str[14:])
d. print(_str[13:])

8) What is the output produced by the following code when the user input is 20.5?

```
try:
    num = int(input("Enter a number: "))
    result = 60 / num
except ZeroDivisionError:
    print("Sorry ! You are dividing by zero ")
except ValueError:
    print("Input Integer value")
finally:
    print('Thanks')
```

a. Thanks
b. Input Integer value
   Thanks
c. Sorry ! You are dividing by zero
   Thanks
d. Nothing

9) Which one of the following statements is used to open a file "temp.txt" so that you can add new lines at the end of the file?

a. outfile = open("temp.txt","w")
b. outfile = open("temp.txt", "rw")
c. outfile = open("temp.txt", "r")
d. outfile = open("temp.txt", "a")

10) What is the use of the **read(n)** function in following code

```
n = 10
data = file.read(n)
```

a. write a string of n characters to a text file
b. read all lines into a list
c. read characters from the nth line only
d. read n number of characters from a file

**Section B - Short Answer Questions (6 Marks)**

| Section B: | / 6 |
|---|---|

11) Read the following code and complete the missing parts in the code. Look at the comments for hints.

```
Patient_dic={"1234":"Ali","4567":"Khamis"}
print(Patient_dic)

#create empty dictionary
new_dic = ……

p_id = input("Enter the Patient id ")
p_name = input("Enter the Patient name ")

#add the input to the Patient_dic
Patient_dic.update(……)

#print all keys and values in the Patient_dic
for …… in ……:
    print(f" {id} and name is {name}")

#remove the item with the key '4567'
……

#copy all elements from Patient_dic
new_dic = ……

```

**Section C - Write the Output**
**(6 Marks)**

| Section C: | / 6 |
|---|---|

12) Refer to the program given below and the file content, write the correct output of the print statement for the specific line number.

```
1  import re
2
3  def file_access(file1):
4          f1 = open(file1,"r")
5          text = f1.readline()
6          print(text)
7          print(text.find('vision'))
8          lst = text.split()
9          print(len(lst))
10         print(lst[3:])
11         lst = re.findall(r'\w+[-]\w+' , text)
12         print(lst)
13         lst = re.findall(r'\d' , text)
14         print(set(lst))
15         f1.close
16
17 file_access("Oman.txt")
```

**Oman.txt**

File Home View

```
Oman vision 2040 is wide-ranging blueprint
for economic and social development.
So Oman will emerge as a prosperous nation.
```

a) The output for the line number 6.

_____

b) The output for the line number 7.

_____

c) The output for the line number 9.

_____

d) The output for the line number 10.

_____

e) The output for the line number 12.

_____

f) The output for the line number 14.

_____

**Section D - Find the Errors**
**(6 Marks)**

| Section D: | / 6 |
|---|---|

**13)** Observe the following Python code very carefully and rewrite it after removing all syntactical errors.

```
def list_convert(t,s):
    try:
        # define a dictionary
        d = {'name':"Ahmed","age":20}
        d.update("gender":"Male")
        # convert all sequences to list
        t_list,s_list,d_list =list(t),list(s),list(d)
        # insert the value 5 after 4 in t_list
        t_list.insert(5)
        print("tuple_to_list: ", t_list)

    finally:
        print("The End")
    except ValueError
        print('Value Error')


t = (4)                # define a tuple
s = 'hello world!'     # define a string
list_convert()         # fun call
```

**Section E - Writing Program Questions**
**(2 x 6 = 12 Marks)**

| Section E: | / 12 |
|---|---|

14) Write a program to solve the following problems. **(6 marks)**

a) Define a function that takes two arguments x and y, and returns the sum of all the numbers between x and y.

b) Using the math module, write a program that prompts the user to enter the side of a hexagon (**s**) and displays its area (**A**).
The formula for computing the area of a hexagon is

$$A = \frac{3 * \sqrt{3}}{2} s^2$$

15) Consider the following tuple to answer the following questions. **(6 marks)**

```
languages = ('Python', 'Swift', 'C++')
```

a) Join the tuple languages with a tuple with one element "Java" and store it back in the same tuple.

_____

b) What is the command to get the output ('C++', 'Python') after adding "Java"?

_____

```
first , *middle, last = languages
```

c) What is the value of middle, after adding "Java"?

_____

d) Duplicate the items in the tuple languages 2 times.

_____

e) Count the number of times 'Python' appears in the tuple.

_____

f) Check if 'C#' exists in the languages tuple.

_____